

# Raspberry Pi Cluster vs. Single Computer for Self-Hosting

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## Abstract

This paper presents a comprehensive analysis comparing two popular approaches to self-hosting: deploying multiple Raspberry Pi single-board computers versus utilizing a single dedicated computer with virtualization technology. We examine multiple dimensions including initial investment costs, operational expenses, performance characteristics, scalability, reliability, and practical use cases. Our analysis includes a detailed breakdown of total cost of ownership (TCO), power consumption patterns, and architectural considerations when using Proxmox VE as a virtualization platform. The findings suggest that while Raspberry Pi clusters offer advantages in learning opportunities and physical redundancy, a single computer with virtualization provides superior performance-per-dollar, simplified management, and greater flexibility for most self-hosting scenarios.

## 1 Introduction

Self-hosting has gained significant popularity among technology enthusiasts, privacy-conscious individuals, and small businesses seeking greater control over their digital infrastructure. Two dominant approaches have emerged: distributed systems using multiple single-board computers (SBCs) like Raspberry Pi, and centralized virtualization using dedicated hardware running platforms such as Proxmox VE.

The choice between these approaches involves trade-offs across multiple dimensions: capital expenditure, operational costs, performance, maintainability, and learning value. This paper provides a quantitative and qualitative analysis to guide decision-making for individuals and small organizations establishing self-hosted infrastructure.

### 1.1 Problem Statement

Given a fixed budget and set of self-hosting requirements, which approach delivers optimal value: multiple Raspberry Pi devices or a single computer with virtualization capabilities?

### 1.2 Scope

This analysis considers:

- Hardware costs for comparable capability levels
- Long-term operational expenses (primarily electricity)
- Performance characteristics and limitations

- Management complexity and time investment
- Scalability and future expansion options
- Redundancy and reliability considerations

## 2 Background

### 2.1 Raspberry Pi Platform

The Raspberry Pi is a series of single-board computers developed by the Raspberry Pi Foundation. The current mainstream model, Raspberry Pi 4 Model B (8GB), features:

- Broadcom BCM2711 quad-core ARM Cortex-A72 CPU @ 1.5GHz
- 8GB LPDDR4 RAM
- Gigabit Ethernet
- USB 3.0 connectivity
- Power consumption: 3-7W typical

The newer Raspberry Pi 5 offers improved performance with BCM2712 quad-core ARM Cortex-A76 @ 2.4GHz, but at a higher price point and power consumption (5-12W).

### 2.2 Dedicated Computer Platform

A dedicated self-hosting computer does not necessarily imply newly built or high-end consumer hardware. In practice, one of the most cost-effective and widely adopted approaches is the use of **used or refurbished enterprise and office computers**.

Such systems are commonly decommissioned by companies after 3–5 years of service despite remaining reliable, energy-efficient, and fully capable of running modern operating systems and virtualization platforms. Due to economies of scale and corporate refresh cycles, these machines are widely available on the secondary market at a fraction of their original cost.

A representative example is a used **Lenovo ThinkCentre M720q Tiny**, acquired for approximately 170€. This system features:

- Intel Core i5-9400T (6 cores, 6 threads, 1.8–3.4 GHz)
- 16GB DDR4 RAM (2666 MHz), expandable to 32GB
- 256GB NVMe SSD (PCIe M.2)
- Integrated Intel UHD Graphics 630
- Gigabit Ethernet, Wi-Fi, and Bluetooth connectivity
- Compact mini-PC form factor with a 65W power adapter

Despite its small physical footprint and low power envelope, this class of system significantly outperforms single-board computers in CPU performance, memory bandwidth, and storage I/O. Additionally, it supports full x86\_64 virtualization with hardware-assisted extensions, making it well suited for platforms such as Proxmox VE.

Used enterprise mini PCs therefore represent an optimal balance between cost, performance, energy efficiency, and reliability for self-hosting workloads.

## 2.3 Proxmox Virtual Environment

Proxmox VE is an open-source virtualization platform combining KVM hypervisor and LXC containers. Key features include:

- Web-based management interface
- High availability clustering
- Built-in backup and restore
- Software-defined networking and storage
- No licensing costs for core features

## 3 Cost Analysis

### 3.1 Initial Capital Expenditure

Table 1 presents a detailed breakdown of initial hardware costs for both approaches. We compare a cluster of 4× Raspberry Pi 4 (8GB) units against a mid-range dedicated computer.

| Table 1: Hardware Cost Comparison (USD, January 2026) |                       |                    |
|-------------------------------------------------------|-----------------------|--------------------|
| Component                                             | Raspberry Pi Cluster  | Single Computer    |
| Main Unit(s)                                          | 4× 75€ = 300€         | -                  |
| CPU                                                   | -                     | 200€(Ryzen 5 5600) |
| Motherboard                                           | -                     | 120€               |
| RAM                                                   | -                     | 270€(32GB DDR4)    |
| Storage (System)                                      | 4× 15€ = 60€(microSD) | 60€(500GB NVMe)    |
| Storage (Data)                                        | 60€(256GB SSD)        | 60€(256GB SSD)     |
| Power Supply                                          | 4× 10€ = 40€          | 60€                |
| Case                                                  | 30€(cluster case)     | 50€                |
| Network Switch                                        | 40€(5-port GB)        | -                  |
| Cooling                                               | -                     | 25€(CPU cooler)    |
| <b>Total</b>                                          | <b>530€</b>           | <b>645€</b>        |

While the Raspberry Pi cluster appears cheaper initially (145€ savings), this calculation assumes used or budget components for the dedicated computer. A new, higher-end configuration could cost 1000-1500€.

### 3.1.1 Used and Refurbished Computer Option

When used or refurbished enterprise hardware is considered, the initial cost comparison changes significantly. Mini PCs such as the Lenovo ThinkCentre, HP EliteDesk, or Dell OptiPlex Micro series are widely available with modern multi-core CPUs, fast NVMe storage, and sufficient memory for virtualization workloads.

Typical specifications in the 150–250€ price range include:

- 6–8 core Intel Core i5 or equivalent processor
- 16GB DDR4 RAM
- NVMe or SATA SSD storage
- Integrated graphics and gigabit networking

At this price point, a used mini PC often costs less than a multi-node Raspberry Pi cluster while delivering substantially higher single-threaded and aggregate performance. As a result, the traditional assumption that Raspberry Pi clusters are the cheapest entry point into self-hosting does not always hold when the secondary market is taken into account.

## 3.2 Operational Costs

Table 2 compares annual electricity costs assuming 24/7 operation at 0.15c /kWh.

Table 2: Annual Power Consumption and Costs

| Configuration               | Avg Power (W) | kWh/year       | Annual Cost        |
|-----------------------------|---------------|----------------|--------------------|
| 4× RPi 4 (idle)             | 20            | 175            | 26.28€             |
| 4× RPi 4 (typical)          | 28            | 245            | 36.79€             |
| Single PC (idle)            | 35            | 307            | 46.01€             |
| Single PC (typical)         | 55            | 482            | 72.27€             |
| Single PC (heavy)           | 85            | 745            | 111.68€            |
| <b>Difference (typical)</b> | <b>27W</b>    | <b>237 kWh</b> | <b>35.48€/year</b> |

## 3.3 Total Cost of Ownership (5 Years)

Considering a 5-year lifecycle:

$$\text{TCO}_{\text{RPi}} = 530\text{€} + (5 \times 36.79\text{€}) = 713.95\text{€}$$

$$\text{TCO}_{\text{PC}} = 645\text{€} + (5 \times 72.27\text{€}) = 1,006.35\text{€}$$

The Raspberry Pi cluster saves approximately 322€ over 5 years, or 64.40€ annually. However, this does not account for potential hardware replacements (SD card failures are common) or performance differences.

## 4 Performance Analysis

### 4.1 Computational Performance

Table 3 compares representative benchmark scores.

| Table 3: Performance Comparison (Normalized Scores) |                 |                     |
|-----------------------------------------------------|-----------------|---------------------|
| Metric                                              | 4× RPi 4 (8GB)  | Ryzen 5 5600 + 32GB |
| Single-core (per unit)                              | 100             | 380                 |
| Multi-core (aggregate)                              | 400             | 2,100               |
| RAM bandwidth (GB/s)                                | 4.8 (each)      | 47.6                |
| Storage I/O (MB/s)                                  | 40-90 (microSD) | 3,500+ (NVMe)       |
| Network (aggregate)                                 | 4 Gbps          | 1-10 Gbps           |

Key observations:

- **Single-thread performance:** The x86\_64 processor offers 3-4× better single-threaded performance, crucial for many applications.
- **Memory:** Centralized system provides 4× total RAM with significantly higher bandwidth.
- **Storage:** NVMe SSD delivers 40-80× faster random I/O compared to microSD cards.
- **Architecture:** x86\_64 compatibility ensures broader software support.

### 4.2 Virtualization Overhead

Proxmox VE running KVM virtual machines introduces 2-5% performance overhead for CPU-bound tasks and 5-10% for I/O operations. LXC containers add less than 2% overhead. This minimal loss is vastly outweighed by the superior base performance.

## 5 Proxmox VE Architecture

Figure 1 illustrates the layered architecture of Proxmox VE, demonstrating how the platform provides both full virtualization and containerization capabilities on a single Linux kernel.

The architecture consists of four distinct layers:

**Linux Kernel Layer:** At the foundation sits the Linux kernel with KVM (Kernel-based Virtual Machine) module for hardware virtualization support, along with AppArmor and cgroups for security and resource management.

**Hypervisor Layer:** QEMU provides the emulation layer for virtual machines, enabling full hardware virtualization with isolated guest operating systems.

**Virtualization Layer:** This layer supports two types of workloads:

- **Virtual Machines (VMs):** Full virtualization running complete guest operating systems (Linux, Windows, BSD) with their own kernels, providing maximum isolation and compatibility

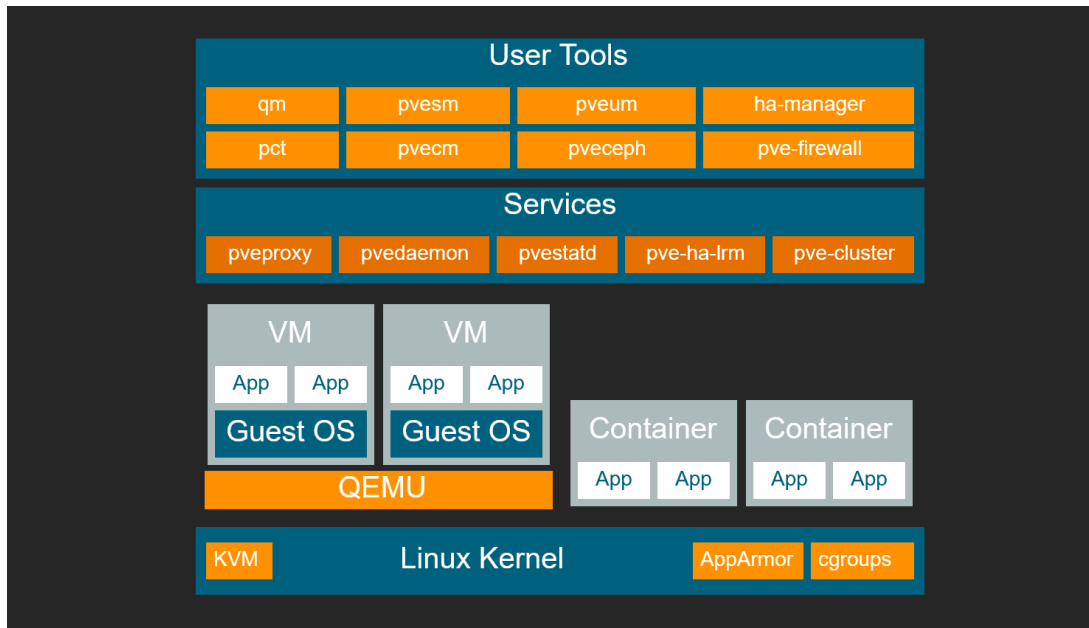


Figure 1: Proxmox VE Architecture Layers

- **Containers:** Lightweight Linux containers sharing the host kernel, offering near-native performance with minimal overhead

**Services Layer:** Core Proxmox services manage the platform:

- **pveproxy:** Web interface and API gateway
- **pvedaemon:** Main management daemon
- **pvestatd:** Statistics collection
- **pve-ha-lrm:** High availability local resource manager
- **pve-cluster:** Cluster communication and configuration

**User Tools Layer:** Command-line and web-based management interfaces:

- **qm:** Virtual machine management
- **pct:** Container management
- **pvesm:** Storage management
- **pveum:** User and permission management
- **pvecm:** Cluster management
- **pveceph:** Ceph storage integration
- **pve-firewall:** Integrated firewall
- **ha-manager:** High availability configuration

This architecture provides isolation comparable to physical separation while maintaining the performance and management benefits of a unified platform. Virtual machines offer complete OS independence, while containers provide lightweight isolation for Linux workloads with minimal performance overhead.

## 6 Comparative Analysis

### 6.1 Advantages of Raspberry Pi Cluster

1. **Physical redundancy:** Hardware failure affects only one service
2. **Lower power consumption:** Approximately 50% less electricity usage
3. **Distributed learning:** Hands-on experience with clustering technologies
4. **ARM architecture exposure:** Valuable for embedded/IoT development
5. **Gradual scaling:** Can add units incrementally as budget allows
6. **Silent operation:** No fans (RPi 4) or quiet fans (RPi 5)

### 6.2 Advantages of Single Computer with Proxmox

1. **Superior performance:** 5-10 $\times$  faster for most workloads
2. **Centralized management:** Single web interface for all services
3. **Better storage options:** NVMe SSDs, RAID configurations, expandability
4. **Software flexibility:** x86\_64 compatibility, Windows VMs possible
5. **Resource efficiency:** Dynamic allocation prevents hardware sitting idle
6. **Simplified networking:** Internal virtual networks, no physical switch needed
7. **Professional relevance:** Proxmox/VMware skills transferable to enterprise
8. **Upgrade path:** Can add RAM, storage, or CPU without replacing everything

### 6.3 Disadvantages Comparison

Table 4: Key Disadvantages of Each Approach

| Raspberry Pi Cluster           | Single Computer                     |
|--------------------------------|-------------------------------------|
| Limited per-node performance   | Single point of failure             |
| SD card reliability issues     | Higher power consumption            |
| ARM compatibility limitations  | Higher initial cost                 |
| Complex distributed management | Requires UPS for safety             |
| Limited RAM per node           | More complex to physically relocate |
| Network congestion possible    | Fan noise (usually minimal)         |

## 7 Use Case Analysis

### 7.1 When Raspberry Pi Cluster is Preferable

- **Educational purposes:** Learning Kubernetes, Docker Swarm, or distributed systems
- **Low-power requirement:** Off-grid or battery-powered scenarios
- **IoT gateway applications:** GPIO pins needed for hardware interfacing
- **Budget constraints with time:** Can build cluster gradually
- **ARM-specific development:** Testing ARM-compiled software

### 7.2 When Single Computer is Preferable

- **General self-hosting:** Web servers, databases, media servers, home automation
- **Performance-critical applications:** Transcoding, compilation, machine learning inference
- **Storage-intensive needs:** NAS functionality, Plex/Jellyfin media library
- **Professional skill development:** Enterprise virtualization experience
- **Windows applications:** Anything requiring x86 Windows VMs
- **Simplified management:** Single system to maintain and update

## 8 Recommendations

Based on our analysis, we provide the following recommendations:

### 8.1 For Most Users: Single Computer with Proxmox VE

This approach offers:

- Better performance-per-dollar ratio
- Simplified management and maintenance
- Greater flexibility for diverse workloads
- Direct transferability to professional environments
- Superior storage and upgrade options

**Recommended specification:**

- CPU: AMD Ryzen 5 5600 or Intel i5-12400
- RAM: 32GB DDR4 (2×16GB for dual-channel)



- Storage: 500GB NVMe (system) + 2TB SSD (data)
- Case: Standard ATX with good airflow
- PSU: 550W 80+ Bronze or better
- UPS: 600-1000VA for graceful shutdown during power loss

## 8.2 For Specific Scenarios: Raspberry Pi Cluster

Choose this approach if:

- Primary goal is learning distributed systems/clustering
- Extremely budget-constrained with time flexibility
- Low-power operation is critical (off-grid, solar)
- ARM development is a specific requirement
- Physical space is severely limited

## 9 Conclusion

While Raspberry Pi clusters offer compelling advantages in power efficiency, physical redundancy, and educational value, a single computer running Proxmox VE provides superior overall value for most self-hosting scenarios. The performance differential is substantial—often 5-10× across key metrics—while the cost premium is modest when amortized over 3-5 years.

The additional 322€ five-year cost of the single computer approach buys:

- Dramatically faster performance for all workloads
- Simplified management through centralized virtualization
- Better storage reliability and expandability
- Professional-grade virtualization experience
- Greater flexibility for future requirements

For users prioritizing learning distributed systems concepts or requiring minimal power consumption, a Raspberry Pi cluster remains a valid choice. However, for general self-hosting purposes—whether home media servers, development environments, or network services—the single computer with Proxmox VE delivers better performance, easier management, and superior long-term value.

The optimal decision ultimately depends on specific requirements, but for most users embarking on self-hosting, we recommend starting with a well-configured single computer. This provides the best foundation for growth while maintaining the option to add Raspberry Pi units later for specialized purposes or redundancy.

## 10 What Can Be Self-Hosted

Modern self-hosting environments support a wide range of applications that traditionally rely on third-party cloud providers. Thanks to containerization, virtualization, and open-source ecosystems, individuals can now operate services that rival commercial SaaS offerings in functionality while retaining full control over their data.

Self-hosted services typically fall into several broad categories: media streaming, networking and security, personal cloud and backup, productivity tools, and infrastructure services.

### 10.1 Media and Entertainment Services

One of the most common entry points into self-hosting is media management and streaming. These workloads benefit greatly from local storage and are well suited for both Raspberry Pi clusters and single-server setups.

**Navidrome** is a lightweight, open-source music streaming server compatible with the Subsonic API. It allows users to stream their personal music library to mobile and desktop clients while maintaining full ownership of the media files. Navidrome has minimal hardware requirements and runs efficiently even on Raspberry Pi hardware, making it an ideal first self-hosted service.

Video streaming platforms such as **Jellyfin** or **Plex** allow users to host personal movie and TV show libraries. These services benefit significantly from higher CPU performance and hardware video acceleration, especially for real-time transcoding, which strongly favors a single x86-based computer over ARM SBCs.

Self-hosting media services eliminates recurring subscription fees, removes geo-restrictions, and ensures long-term access to personal content regardless of external provider policies.

### 10.2 Networking, Security, and Privacy Services

Self-hosting is particularly attractive for network-level services that enhance privacy, security, and reliability.

**AdGuard Home** is a DNS-based ad and tracker blocking solution that operates at the network level. By filtering DNS queries, it prevents advertisements and malicious domains from loading on all connected devices without requiring client-side software. Running such a service locally improves privacy, reduces bandwidth usage, and avoids reliance on external DNS providers.

Other commonly self-hosted networking services include:

- **Unbound or Pi-hole** for recursive DNS resolution
- **WireGuard or OpenVPN** for secure remote access
- **Reverse proxies** such as Nginx or Traefik for HTTPS termination
- **Firewalls and monitoring tools** for traffic analysis

These services are generally lightweight and well suited for Raspberry Pi devices, but centralizing them on a Proxmox host simplifies management and enables redundancy via virtual machines or containers.

## 10.3 Personal Cloud, Photos, and Backup Services

Another major category of self-hosted applications replaces commercial cloud storage and photo management platforms.

**Immich** is a modern, open-source photo and video backup solution designed as a self-hosted alternative to Google Photos. It provides automatic uploads from mobile devices, facial recognition, metadata indexing, and fast browsing of large media libraries. Due to its reliance on databases, background processing, and optional machine learning features, Immich benefits significantly from fast storage and higher memory availability.

**Nextcloud** and similar platforms provide file synchronization, document editing, calendar, and contact management services. These systems centralize personal data while allowing access from multiple devices without surrendering control to external cloud providers.

Backup solutions such as **BorgBackup**, **Restic**, or **Proxmox Backup Server** are commonly self-hosted to ensure data integrity and disaster recovery without ongoing subscription costs.

## 10.4 Infrastructure and Developer Services

Self-hosting is widely used by developers and small teams to operate internal infrastructure services.

Common examples include:

- **Git repositories** (Gitea, GitLab)
- **CI/CD systems** (GitLab Runner, Jenkins)
- **Databases** (PostgreSQL, MySQL, Redis)
- **Monitoring and logging** (Prometheus, Grafana, Loki)
- **Container platforms** (Docker, Kubernetes)

These services benefit from centralized resource management, fast storage, and sufficient memory, making a single computer with virtualization particularly well suited. Proxmox VE allows isolation between services while sharing hardware efficiently.

# 11 General Advantages of Self-Hosting

Self-hosting provides several strategic advantages over reliance on third-party cloud and SaaS providers.

## 11.1 Data Ownership and Privacy

All data remains under the physical and administrative control of the owner. This eliminates concerns related to data mining, provider policy changes, vendor lock-in, or service shutdowns. For privacy-conscious users, this is often the primary motivation.

## 11.2 Cost Predictability

While self-hosting requires upfront investment, ongoing costs are largely limited to electricity and occasional hardware upgrades. Over time, this can be significantly cheaper than recurring subscription fees for multiple cloud services.

## 11.3 Customization and Flexibility

Self-hosted services can be configured, extended, or modified without restrictions imposed by commercial providers. Users are free to integrate services, automate workflows, and experiment with new technologies.

## 11.4 Learning and Skill Development

Operating self-hosted infrastructure provides hands-on experience with Linux, networking, virtualization, security, and DevOps practices. These skills are directly transferable to professional IT and software engineering roles.

## 11.5 Resilience and Independence

Self-hosting reduces dependency on external providers and internet availability. Local services continue functioning during outages, and backups can be managed according to individual risk tolerance rather than provider policies.

# 12 Configuration Management and Automation with Ansible

As self-hosted environments grow in complexity, manual configuration and maintenance quickly become error-prone and time-consuming. Configuration management tools address this problem by enabling repeatable, automated, and declarative infrastructure management. One of the most widely adopted tools in this space is **Ansible**.

Ansible is an open-source automation platform designed for configuration management, application deployment, and infrastructure orchestration. It is particularly well suited for small to medium-scale self-hosting environments due to its simplicity, agentless architecture, and strong community support.

## 12.1 Core Concepts of Ansible

Ansible operates over SSH and does not require any agent software to be installed on managed nodes. This makes it ideal for environments consisting of heterogeneous devices such as Raspberry Pi clusters and x86-based servers.

Key concepts include:

- **Inventory:** A list of managed hosts, which can be static or dynamically generated
- **Playbooks:** YAML-based files describing desired system state
- **Roles:** Reusable collections of tasks, templates, and variables

- **Modules:** Pre-built units of work (e.g., package installation, service management)
- **Idempotency:** Ensures repeated execution produces the same system state

Idempotency is particularly important in self-hosted systems, as it allows safe re-application of configurations after failures, reboots, or hardware replacements.

## 12.2 Automating Self-Hosted Infrastructure

Ansible can automate nearly every layer of a self-hosted stack, from base operating system configuration to application deployment.

Typical automation tasks include:

- Operating system updates and package installation
- User and SSH key management
- Firewall and network configuration
- Docker and container runtime installation
- Deployment of services such as Navidrome, AdGuard Home, and Immich
- Configuration of reverse proxies and TLS certificates
- Scheduled backups and monitoring agents

By encoding these steps into playbooks, infrastructure becomes **self-documenting**. A complete environment can be recreated from scratch on new hardware with minimal manual intervention.

## 12.3 Ansible in Raspberry Pi Clusters

In Raspberry Pi clusters, Ansible significantly reduces operational complexity. Instead of manually configuring each node, administrators can define node roles such as *DNS server*, *media server*, or *worker node*, and apply configurations uniformly.

Common use cases include:

- Consistent OS hardening across all nodes
- Automated Docker Swarm or Kubernetes node setup
- Rolling updates to avoid service downtime
- Rapid recovery after SD card or node failure

This approach mitigates one of the primary disadvantages of Raspberry Pi clusters: the increased management overhead associated with distributed systems.

## 12.4 Ansible in Virtualized Environments

When combined with Proxmox VE, Ansible enables full lifecycle automation of virtual machines and containers. While Proxmox handles virtualization and resource allocation, Ansible manages software configuration inside each guest.

Typical workflows include:

- Provisioning new virtual machines from templates
- Automated post-install configuration of Linux guests
- Service deployment via Docker Compose
- Coordinated updates across multiple services

This separation of concerns allows administrators to treat virtual machines as disposable resources. If a system becomes misconfigured, it can be destroyed and recreated using the same Ansible playbooks, ensuring consistency and reducing downtime.

## 12.5 Benefits of Ansible for Self-Hosting

The use of Ansible in self-hosted environments provides several key benefits:

- **Consistency:** All systems converge toward a known, tested configuration
- **Scalability:** Adding new services or hosts requires minimal additional effort
- **Disaster recovery:** Infrastructure can be rebuilt quickly after hardware failure
- **Reduced human error:** Manual, repetitive tasks are eliminated
- **Version control:** Infrastructure definitions can be stored in Git
- **Documentation:** Playbooks serve as executable documentation

For individual users and small teams, Ansible bridges the gap between ad-hoc system administration and professional infrastructure-as-code practices.

## 12.6 Limitations and Considerations

While Ansible is well suited for self-hosting, it is not without limitations. Initial setup requires familiarity with YAML and Linux administration concepts. Additionally, Ansible operates in a push-based model, which may not be ideal for highly dynamic or ephemeral environments without additional tooling.

Despite these limitations, the benefits of automation generally outweigh the learning curve, especially as infrastructure size and service count increase.

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